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DEPARTMENT OF MATHEMATICAL SCIENCES

(Faculty of Science)

SA'ADU ZUNGUR UNIVERSITY GADAU

End of Second Semester Examination 2024/2025 Session

Course Code and Title: CSC2204/IFT212/CSC204 (COMPUTER ARCHITECTURE AND ORGANIZATION) Time Allowed: (2 hours)

Credit Units: (2 Units)

Instruction: Answer any four (4) questions

Q1.

- a). Discuss three (3) types of addressing modes you know. (6marks)
- b). List any four (4) registers available in the CPU and their function. (4marks)
- c). Explain Addressing Mode. (5marks)

Q2.

- a). List and briefly discuss the type of CPU. (3marks)
- b). Briefly discuss the function of the three (3) major building blocks of the CPU. (6marks)
- c). Discuss any three (3) instruction types. (6marks)

Q3.

- a). Explain the following with example: (10marks)
 - i. Three address instruction format
 - ii. Two address instruction format
 - iii. One and half address instruction format
 - iv. One address instruction format
 - v. Zero address instruction format

Q4.

- a). Discuss the CPU Instruction Execution Cycle. (5marks)
- b). Discuss the term cache hit and cache miss related to cache memory. (4marks)
- c). Differentiate between SRAM and DRAM. (5marks)

Q5.

- a). With the aid of a diagram, discuss memory hierarchy. (6marks)
- b). List and explain three (3) characteristics of memory hierarchy. (6marks)
- c). Discuss virtual memory. (5marks)
- d). Explain the following term: (4marks)
 - i. Paging
 - ii. Segmentation

GOOD LUCK